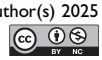
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Social Media Advertising's Role in Small Business Resilience Amid Lockdowns

Shun-Yang Lee, Julian Runge, Daniel Yoo, Yakov Bart, Anett Gyurak, and J.W. Schneider

Abstract

Research has investigated the impact of the COVID-19 pandemic on business performance and survival, indicating particularly adverse effects for small and midsize businesses (SMBs). Yet only limited work has examined whether and how advertising may have helped shape these outcomes, particularly for SMBs. By constructing and analyzing a novel dataset of more than 79,000 businesses across 49 countries, the authors explore this question and examine the impact of pandemic-induced government stay-at-home orders on business survival. Using staggered difference-in-differences estimators, this research produces directional evidence that government lockdowns increased the likelihood of SMB closure and that social media advertising attenuated this adverse effect. These results are robust across different model specifications and measures of advertising activities. The findings additionally show heterogeneity across business sizes and industries, enriching the understanding of business survival during severe demand crises from a global perspective. The current research suggests that advertising may play an important role in building SMB resilience in light of severe demand shocks and mobility restrictions.

Keywords

small business, social media advertising, COVID-19, lockdown, stay-at-home, business closure

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The COVID-19 pandemic posed a serious threat to the operation of businesses around the world. As the pandemic unfolded, businesses were faced with the dire consequences of falling demand, declining revenues, and liquidity constraints. Indeed, as the pandemic intensified, businesses closed at an alarming rate (Apedo-Amah et al. 2020).[1] While the pandemic has increased researchers' interest in understanding the extent and moderators of these closures (Bartik et al. 2020a; Bartlett and Morse 2020; Gourinchas et al. 2025; Wang et al. 2020), few insights exist into how advertising may have shaped the performance and operation of businesses around the world.

One reason for the lack of insight on the interaction of COVID-19 demand shocks, business operation, and advertising is the absence of a guiding theoretical framework for understanding whether/how advertising could help businesses operate in times of unexpected demand crises. While prior studies have examined how business outcomes, such as revenues (Bloom, Fletcher, and Yeh 2021; Chetty et al. 2024), layoffs (Bartik et al. 2020a; Khamis et al. 2021), and business closures (Apedo-Amah et al. 2020; Crane et al. 2022), were affected during the pandemic, few studies have examined the extent to which advertising shaped these outcomes. Furthermore, advertising technology such as digital marketing channels can be particularly important for small and midsize businesses (SMBs) during such crises, as SMBs often mainly rely on digital channels for advertising; however, the literature is lacking in understanding whether/how such advertising technologies can affect SMB operations during demand crises.

Another reason for the lack of research is the paucity of data that observe both the advertising and operation of businesses

[1] Apedo-Amah et al. (2020) find that the likelihood of business closures within four weeks of the peak of the COVID-19 pandemic around the world was more than 70%.

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during crises, such as the pandemic. Although there has been a critical need to understand how businesses were impacted by the pandemic, the sudden and rapid onset of COVID-19 has made it difficult to obtain timely and detailed data, especially outside a small group of regions and countries. In addition, recent efforts have focused on collecting data on a basic set of financial and employment indicators (Apedo-Amah et al. 2020; Bartik et al. 2020b; Chetty et al. 2024). Thus, the availability of advertising data has been limited by comparison, and, to the authors' best knowledge, no studies have closely examined the role of advertising in business survival during the pandemic.

Therefore, in this research, we examine whether advertising impacted SMBs' operations during the COVID-19 pandemic. Consisting of more than 79,000 businesses across 49 countries, our dataset combines firm-level time series of advertising activities on the Facebook platform with information on business operation and data on government lockdowns. We first investigate how governments' stay-at-home orders affected the survival of businesses. We then explore whether Facebook advertising moderated the impact of stay-at-home orders on business operation and potential heterogeneity in the moderating effect. Our approach is exploratory in nature and focuses on providing directional evidence of advertising's potential positive effects on SMBs during severe demand downturns.

Our empirical analyses utilize staggered difference-in-differences (DID) estimators. We find that government lockdowns during the pandemic led to an increase in business closures, but that social media advertising attenuated the impact of these lockdowns. These results are robust across different model specifications and measures of advertising activities. Our findings provide several important implications. First, we confirm that stay-at-home orders have a negative effect on business operation. Second, we find that advertising played an important role in supporting SMBs in navigating the hardships induced by the pandemic and related lockdowns. Third, we uncover effect heterogeneity across business sizes and industries using a global sample. While prior studies on the impact of stay-at-home orders have largely focused on a single country or region (Alexander and Karger 2023; Aum, Lee, and Shin 2021; Bartlett and Morse 2020; Goolsbee and Syverson 2021), our findings highlight the important role that advertising can play in building small business resilience using a global sample.

Literature Review

The COVID-19 pandemic has led to a growing body of research examining how businesses have been affected around the world. Notably, the role of government lockdowns and their impact on the performance and survival of businesses have been a major topic of interest. Alexander and Karger (2023) and Gourinchas et al. (2025) find that stay-at-home orders caused considerable reductions in consumer spending and led to increased failures among small businesses during the pandemic, respectively. Apedo-Amah et al. (2020) also find that business closures around the world were at their highest in the weeks following the most stringent periods of lockdown.

While research on business outcomes during the COVID-19 pandemic has grown rapidly since its onset, several important questions remain unanswered. First, how did advertising influence businesses during the pandemic? Second, to what extent are the results heterogeneous across different types of business? We expect advertising to have played a critical role in the performance and survival of businesses during the pandemic for both theoretical and empirical reasons. In particular, marketing theory posits that advertising activities have a long-term effect (Breuer and Brettel 2012; Reimer, Rutz, and Pauwels 2014) and that they can have a positive effect on future cash flows by supporting the development of market-based intangible assets (McAlister et al. 2016; Srivastava, Shervani, and Fahey 1998). Studies have also shown that advertising has an important influence on business survival, with past advertising both reducing the risk of bankruptcy (Jindal and McAlister 2015) and increasing the likelihood of surviving the bankruptcy process (Jindal 2020). In addition, studies have shown that personalized online advertising can enhance market-based intangible assets, including brand equity, by enhancing brand identification (Gordon, Moakler, and Zettelmeyer 2023; Tran et al. 2020). Personalized online advertising also has a positive impact on brand-specific page views (Rutz and Bucklin 2012), active and passive searches on a business's web page (Ghose and Todri-Adamopoulos 2016), and both revenues and repeat purchase probabilities for businesses (Lewis and Reiley 2014; Manchanda et al. 2006). Moreover, Alekseev et al. (2023) provide initial evidence that Facebook advertising may have mitigated the adverse effect of the pandemic on businesses.[2] Given its ability to target and provide personalized content, we expect social media advertising to be effective in helping businesses accumulate market-based assets that support their financial performance during lockdowns, thereby increasing their likelihood of survival.

Therefore, this article aims to address these questions and offer new insights into the role of advertising for businesses during the pandemic, using a global sample.

Data and Measures

Measuring Business Survival

We obtained data on business survival during the COVID-19 pandemic from the Future of Business Survey (FOBS). The FOBS is a joint data collection project conducted by the World Bank, the Organisation for Economic Co-operation and Development, and Facebook that collects survey data on the challenges, opportunities, and needs of businesses around the world using the Facebook platform. We use the FOBS data to construct an outcome variable of business operation by calendar month. Specifically, we construct panel data consisting of businesses and their monthly (binary) closure decision from December 2019 to May 2020. Tables 1 and 2 provide summary statistics for main continuous and categorical variables, respectively, broken down by whether the business ran any Facebook advertising campaign.

[2] Note that Alekseev et al. (2023) focus on small businesses in the United States, whereas the current research considers a global sample.

Table 1. Summary Statistics by Advertiser Status.

	Advertisers (N = 51,977)				Nonadvertisers (N = 27,140)			
	Mean	SD	Min	Max	Mean	SD	Min	Max
No. of active ads	4.26	13.94	.00	1,192.47	.00	.00	.00	.00
Log(ad spend)	7.81	1.62	.00	16.39	.00	.00	.00	.00
Log(ad spend industry mean)	9.62	1.11	6.54	15.05	9.46	1.14	6.54	15.05
Log(no. of Facebook fans)	6.72	1.80	.00	16.52	5.77	2.06	.00	16.79
Has closed	.12	.32	.00	1.00	.15	.36	.00	1.00
First treatment period	3.83	.77	.00	5.00	3.85	.80	.00	5.00
Is top 50% advertiser	.93	.26	.00	1.00	.00	.00	.00	.00

Notes: First treatment period = 0 represents businesses that were never treated. First treatment period = 1 represents December 2019, 2 represents January 2020, and so forth. Top 50% advertisers are those with an above-median number of active ads.

Table 2. Business Size and Industry by Advertiser Status.

	Advertisers		Nonadvertisers	
	N	Percentage	N	Percentage
Size				
<10 employees	43,388	54.84	22,503	28.44
10–49 employees	6,242	7.89	3,011	3.81
50–249 employees	1,376	1.74	797	1.01
250–499 employees	249	.31	173	.22
≥500 employees	722	.91	656	.83
Industry				
Agriculture	901	1.14	768	.97
Construction	2,044	2.58	1,345	1.70
Hotels	4,477	5.66	2,513	3.18
Information	4,981	6.30	2,432	3.07
Manufacturing	2,047	2.59	1,281	1.62
Other	8,941	11.30	4,449	5.62
Retail	11,443	14.46	5,927	7.49
Services	16,336	20.65	7,838	9.91
Transportation	807	1.02	587	.74
All	51,977	65.70	27,140	34.30

Notes: Size is determined by number of employees.

Measuring Government Responses to the COVID-19 Pandemic

We obtain data on government responses to the COVID-19 pandemic from the Oxford COVID-19 Government Response Tracker, a data collection project that gathered information on the policies and interventions that governments around the world adopted in response to the pandemic. Our primary government response variable is the implementation of stay-at-home orders, which we treat as a binary measure. This variable captures the effects of lockdowns and movement restrictions. Operationally, we consolidate daily records of stay-at-home orders into monthly observations. We classify a month as having a stay-at-home order in effect if the order was implemented for at least one day during the focal month.

Measuring Advertising Activities

We obtained advertising data from Facebook's advertising database. Advertising on Facebook enables businesses to reach consumers who they believe will be most interested in their ads according to a variety of behavioral and demographic characteristics with highly personalized messages. A key benefit of using this data source is that it enables us to observe the actual advertising behavior for businesses in our sample as opposed to self-reported advertising behavior. We collected data on the

advertising use and spending history of businesses in the FOBS sample for our empirical analyses.

Method

We use the collected data to examine how SMBs responded to governments' stay-at-home orders during the COVID-19 pandemic and whether this relationship is moderated by their Facebook advertising activities. Since business closure is a binary outcome (i.e., open or close), we use a logit model for our analyses. Moreover, since some countries never issued any stay-at-home orders, we could use a DID approach to estimate the effect of the stay-at-home order on business closure, where we consider governments' stay-at-home orders as shocks and evaluate business closure as the outcome. However, since different countries issued the stay-at-home orders at different times, this means that the orders were enforced in a staggered fashion across countries. We therefore adopt a staggered DID model for our analysis. Specifically, we adopt an extended two-way fixed effect (ETWFE) approach (Berman and Israeli 2022; Wooldridge 2021, 2023) to model the staggered policy implementation. Intuitively, the ETWFE approach extends the two-way fixed effect model by adding additional interaction terms, including those between treatment and periods as well as between treatment and cohorts, where each cohort includes countries that adopted the treatment in the same period. The advantage of ETWFE is that it addresses the staggered adoption concern and can be applied to nonlinear models, and therefore we use ETWFE as our main modeling approach. As a robustness check, we also implement and present results from using the Callaway and Sant'Anna DID estimator (Callaway and Sant'Anna 2021). Additionally, we use our models to examine whether advertising activities on Facebook moderate the impact of governments' stay-at-home orders. Operationally, we estimate the staggered DID models separately for advertisers and nonadvertisers, enabling us to compare their respective treatment effect estimates.

Model-Free Evidence

We begin by providing model-free evidence of the effect of stay-at-home orders on business closure. We visualize the proportion of businesses that closed separately for advertisers and nonadvertisers in Figure 1. We consider two ways of measuring time: calendar month and number of months since the implementation of stay-at-home orders (i.e., a relative time scale). We find that with both time measurements, advertisers exhibited lower closing proportions than nonadvertisers. Specifically, the relative time plot shows that the proportion of businesses that closed increased drastically as soon as stay-at-home orders took effect (i.e., when Months Since Treatment = 0), suggesting that stay-at-home orders are positively correlated with business closure. We also observe that advertisers exhibited a lower closing rate compared with nonadvertisers across different industries, as shown in Figure 2, and for small businesses (i.e., fewer than 500 employees), as shown in Figure 3. Overall, these findings suggest that advertising might have moderated the effect of stay-at-home orders. We next present results from staggered DID analyses.

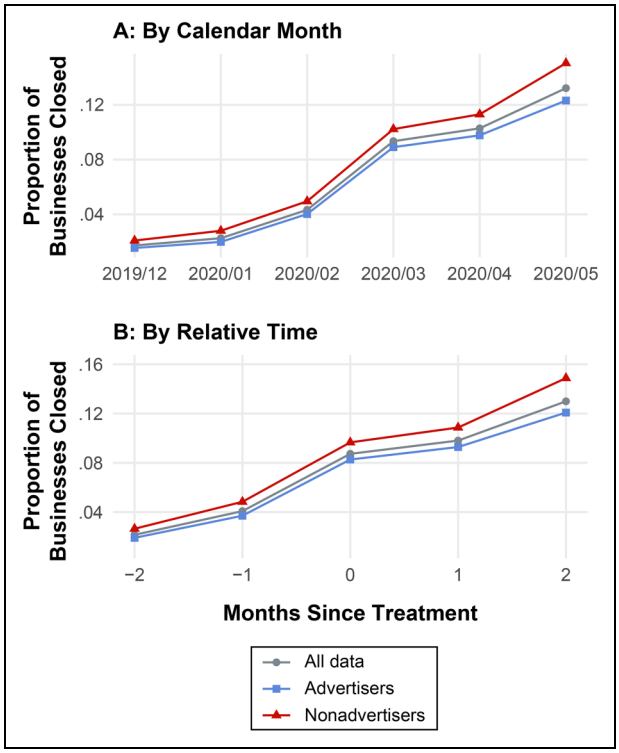


Figure 1. Proportion of Business Closures by Advertiser Status.
Notes: The upper graph displays the closure proportion by calendar month, while the lower graph shows the closure proportion by relative time, with Months Since Treatment = 0 indicating the month in which the stay-at-home order went into effect.

Modeling Results

Main Effect of Stay-at-Home Orders on Business Closure

As described in the "Method" section, our main empirical model is an ETWFE with a logit link to evaluate the effect of stay-at-home orders on a binary outcome (i.e., business closure). We report the estimate of the average treatment effect on the treated (ATT) in Table 3, Column 1. The logit model result shows that the stay-at-home order leads to a 2.9 percentage point increase in business closure probability. We further report the event-study estimates of ATT in Table 3, Column 2, where we define a given period using the number of months since the treatment took effect. For example, ATT (0) represents the estimate for the period when treatment took effect. We find that the ATT estimates kept increasing post treatment. As a robustness check, in Table 3, Columns 3 and 4, we report a set of linear probability model (LPM) results of both the ATT and its event-study estimates, respectively. The LPM results are largely consistent with the logit results. Overall, we find that, consistent with

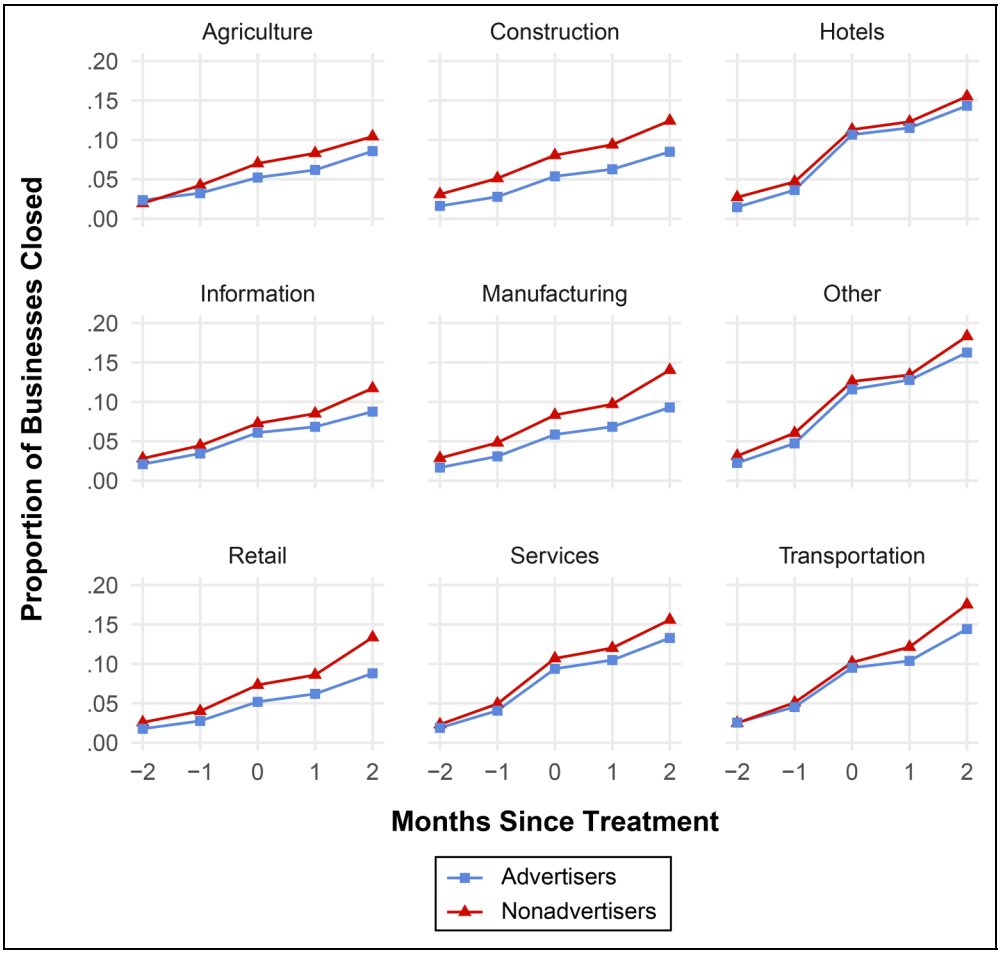


Figure 2. Proportion of Business Closures by Industry and Advertiser Status.
Notes: Each subplot presents the closure proportion over relative time for a specific industry, where Months Since Treatment = 0 denotes the month in which the stay-at-home order was implemented.

expectations, the stay-at-home order is positively correlated with an increase in business closure probability.

Advertising's Moderating Effects

To investigate advertising's moderating effects, we examine the ATT separately for advertisers and nonadvertisers. We measure a business's Facebook advertiser status using two different measures because advertising is known to have a lagged effect (Beaver, McNichols, and Rhie 2005; Jindal and McAlister 2015; Shumway 2001). The first measure represents a longer time frame, that is, using a business's advertising activities in 2019, whereas the second measure represents a shorter time frame, that is, using the advertising activities two months prior to a given period. The results using both measures are reported in Table 4. We also visualize the ATT estimates for the 2019 data in Figure 4 and those for the two-month-lead data in Figure 5. We find that, while the ATT estimates are positive for both advertisers and nonadvertisers across all periods, the advertisers' estimates are consistently lower than those of nonadvertisers. This provides evidence that advertising might have attenuated the negative effect of stay-at-home orders.[3,4]

Alternative Modeling Framework

As a robustness check, we report the event-study type ATT results using the Callaway and Sant'Anna DID estimator (Callaway and Sant'Anna 2021), which we visualize in Figure 7 and display in Table 6. Column 1 shows the ATT estimates for the entire dataset, while Columns 2 and 3 report subgroup analyses results separately for advertisers and nonadvertisers, respectively. We again find that the ATT estimates for nonadvertisers are consistently larger than those of advertisers, consistent with the

[3] We additionally conduct the ETWFE analysis including business fixed effects. Note that by including business fixed effects, we lose potentially relevant within-country variations. We report the results in Table 5. The ATT estimates are consistent with the main results, while the results of the event-study estimates are mixed.
[4] We consider an alternative classification in which we classify a month as having a stay-at-home order in effect if the order was implemented for at least two days during the focal month. The regression results are almost identical and are displayed in Figure 6.

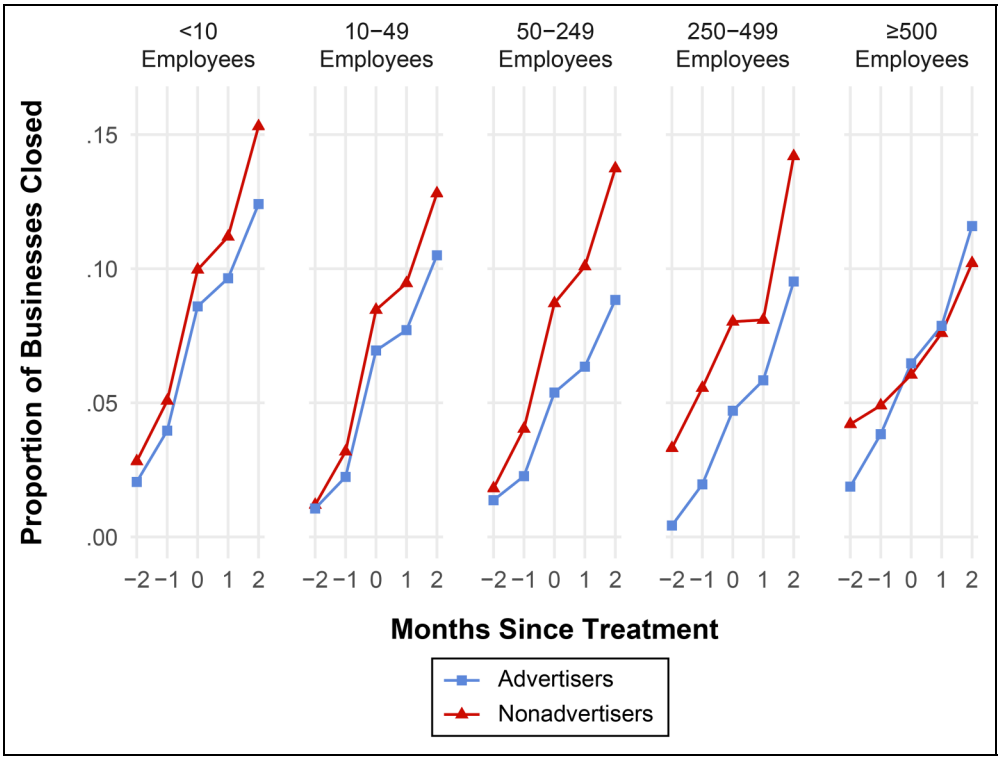


Figure 3. Proportion of Business Closures by Size and Advertiser Status.
Notes: Each subplot displays the closure proportion over relative time for a specific business size category, where size is defined by the number of employees and is indicated above each subplot.

Table 3. Effect of Stay-at-Home Order on Business Closure Probability (ETWFE).

	Logit (ATT)	Logit (Event)	LPM (ATT)	LPM (Event)
ATT	.029 (.008)***		.036 (.007)***	
ATT (0)		.022 (.009)**		.032 (.010)**
ATT (1)		.027 (.007)***		.030 (.006)***
ATT (2)		.035 (.009)***		.041 (.008)***
ATT (3)		.038 (.007)***		.050 (.002)***
ATT (4)		.115 (.017)***		.151 (.003)***
R^2	.080	.080	.041	.041

**$p < .01$.
***$p < .001$.
Notes: The table shows ETWFE (Wooldridge 2021) estimates of ATT, controlling for country gross domestic product (GDP) per capita, business size, and social media popularity (number of Facebook fans). Treatment is the implementation of a stay-at-home order. Outcome is the treatment effect on business closure probability. ATT (Period) refers to the number of months since the treatment took effect (e.g., Period = 2 means two months after the treatment took effect). ETWFE mechanically sets coefficients of pretreatment periods to 0, and thus these coefficients are omitted in the table. Standard errors are clustered at the country level.

ETWFE estimates reported previously. We also notice that the ATT estimate for period −1 (i.e., one month prior to the implementation of the stay-at-home order) is significant for the overall data and for the nonadvertiser subgroup. This suggests there might have been an anticipation effect, causing more businesses to close even before the stay-at-home order took effect. Nevertheless, this appears to be mostly driven by nonadvertisers.[5]

[5] Note that the ETWFE model mechanically sets all pretreatment period ATT estimates to 0, and therefore we are unable to assess the pretreatment trends in the main model.

Alternative Advertising Measures

Recall that in the preceding analyses, we used advertiser status to categorize businesses into advertisers and nonadvertisers. We explore three alternative measures of advertising activities. First, we use the number of ads published to categorize businesses into two halves (i.e., top 50% and bottom 50%) using the median number of active Facebook ads as the threshold. We repeat the ETWFE logit analysis for the two subgroups and report the results in Table 7 and Figure 8. The results again demonstrate that active advertisers had a lower ATT than nonadvertisers.

Table 4. Moderating Effect of Advertising on ATT (ETWFE-Logit).

	Advertiser Status		Advertiser Status (Two-Month Lead)	
	Nonadvertisers	**Advertisers**	**Nonadvertisers**	**Advertisers**
ATT (0)	.026 (.009)**	.021 (.009)*	.037 (.013)**	.035 (.011)**
ATT (1)	.032 (.008)***	.026 (.008)***	.047 (.012)***	.045 (.010)***
ATT (2)	.039 (.010)***	.030 (.009)***	.067 (.015)***	.059 (.012)***
ATT (3)	.051 (.009)***	.033 (.007)***	.080 (.012)***	.062 (.009)***
ATT (4)	.121 (.018)***	.109 (.017)***	.098 (.020)***	.093 (.018)***

*$p < .05$.
**$p < .01$.
***$p < .001$.
Notes: The table shows ETWFE (Wooldridge 2021) estimates of ATT by advertiser status, controlling for country GDP per capita, business size, and social media popularity (number of Facebook fans). The first two columns report results using advertising data from the first half of 2019 to determine advertiser status, while the last two columns report results using advertising data from the two months preceding each period. Standard errors are clustered at the country level.

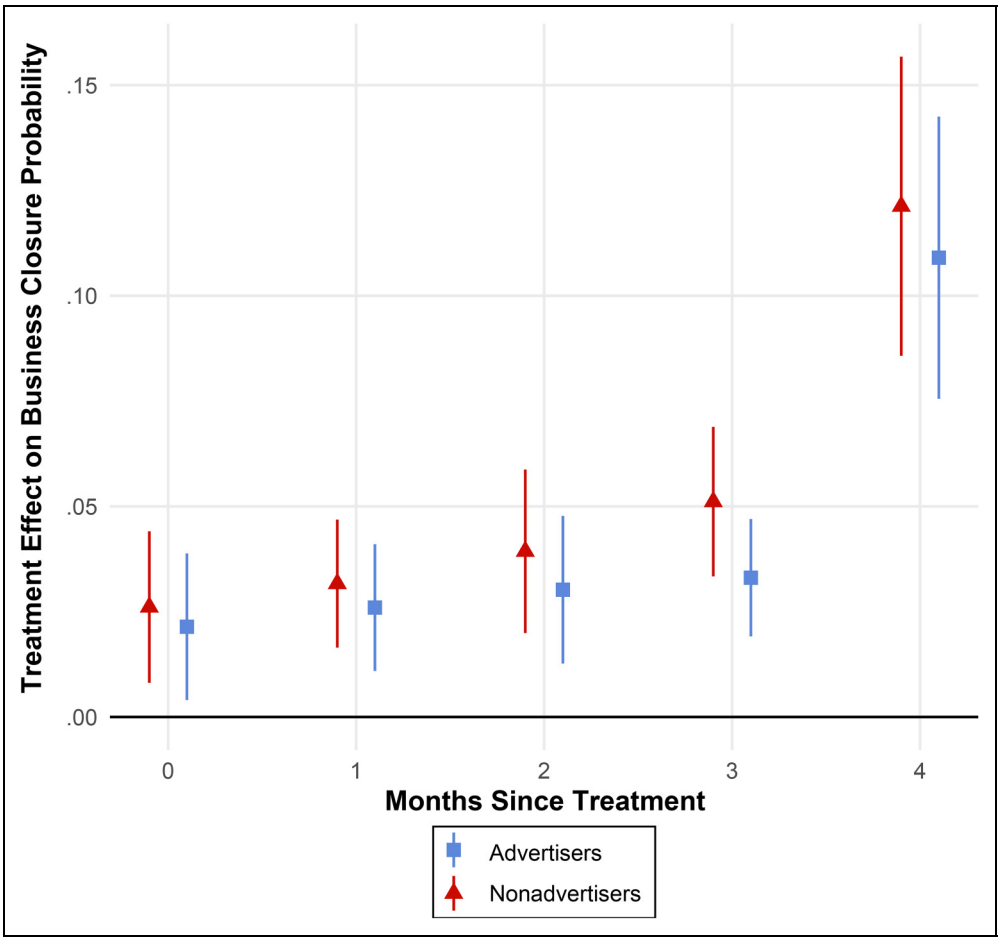


Figure 4. ATT Event-Study Estimates by Advertiser Status.
Notes: The treatment is defined as the implementation of a stay-at-home order. Estimates are derived from an ETWFE-logit model.

Second, among advertisers, we use their ad spending amounts to categorize businesses into five quintiles, with Quintile 1 being the smallest and Quintile 5 the largest in terms of ad spending. We then compute the ATT estimates for the five quintiles separately and report the ETWFE results in Table 8, Row 1. We find that businesses with higher ad spending had lower ATT estimates than those with lower ad spending.

Third, businesses' advertising decisions could be driven by some unobserved factors, leading to endogeneity concerns. To this end, we construct another advertising measure using the

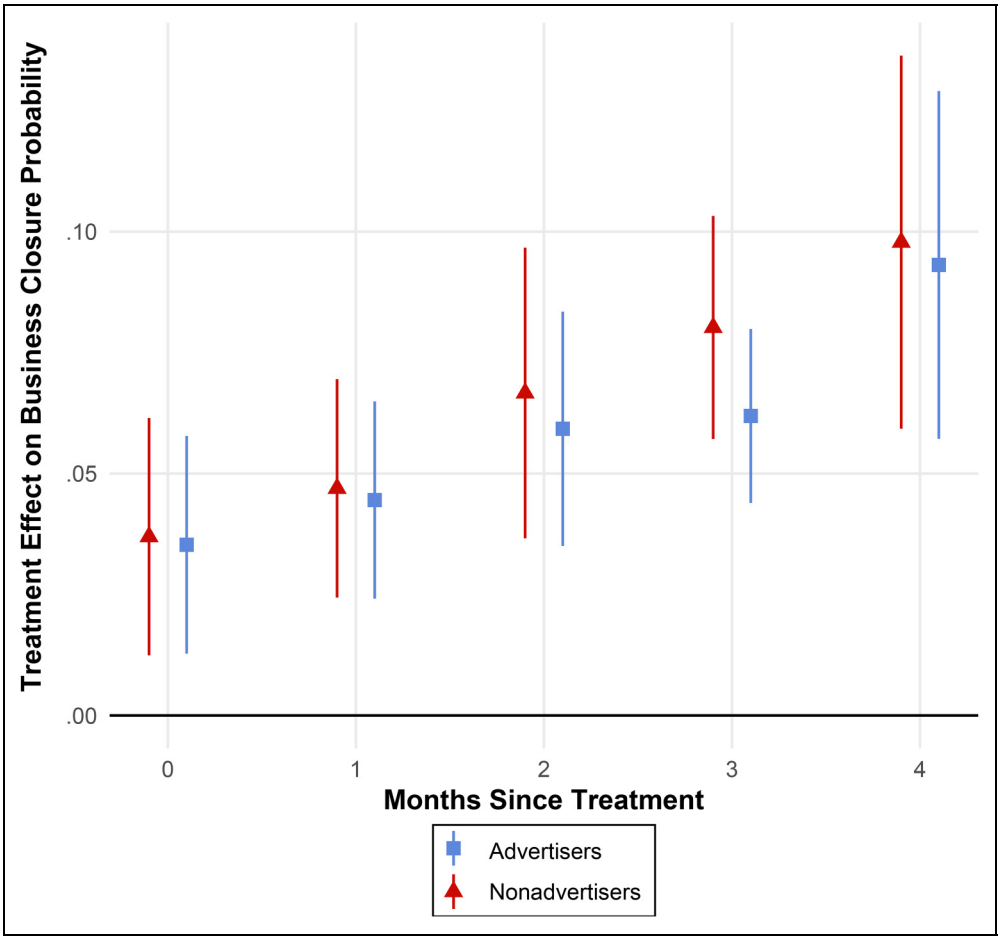


Figure 5. ATT Event-Study Estimates by Advertiser Status with Two-Month Lead.
Notes: Advertising data are based on a two-month lead. The treatment is defined as the implementation of a stay-at-home order. Estimates are derived from an ETWFE-logit model.

Table 5. Effect of Stay-at-Home Order on Business Closure Probability (ETWFE) with Business Fixed Effects.

	Logit (ATT)		Logit (Event)	
	Nonadvertisers	**Advertisers**	**Nonadvertisers**	**Advertisers**
ATT	.034 (.011)**	.030 (.008)***		
ATT (0)			.018 (.016)	.023 (.009)*
ATT (1)			.033 (.013)*	.029 (.007)***
ATT (2)			.047 (.011)***	.034 (.009)***
ATT (3)			.039 (.014)**	.041 (.007)***
ATT (4)			.111 (.015)***	.122 (.012)***

*$p < .05$.
**$p < .01$.
***$p < .001$.
Notes: The table shows ETWFE (Wooldridge 2021) estimates of ATT, controlling for business fixed effects. Treatment is the implementation of a stay-at-home order. Outcome is the treatment effect on business closure probability. ATT (Period) refers to the number of months since the treatment took effect (e.g., Period = 2 means two months after the treatment took effect). ETWFE mechanically sets coefficients of pretreatment periods to 0, and thus these coefficients are omitted in the table. Standard errors are clustered at the country level.

industry average of ad spending. Specifically, for a focal business, we measure its ad spending using the average spending of its competitors in the same industry and country, excluding its own expenditures. The advantage of this approach is that, while the industry average is likely correlated with the focal business's ad spending, it will likely be uncorrelated with the unobserved factors unique to the focal business, alleviating the endogeneity concerns. We similarly categorize businesses into five quintiles based on the industry

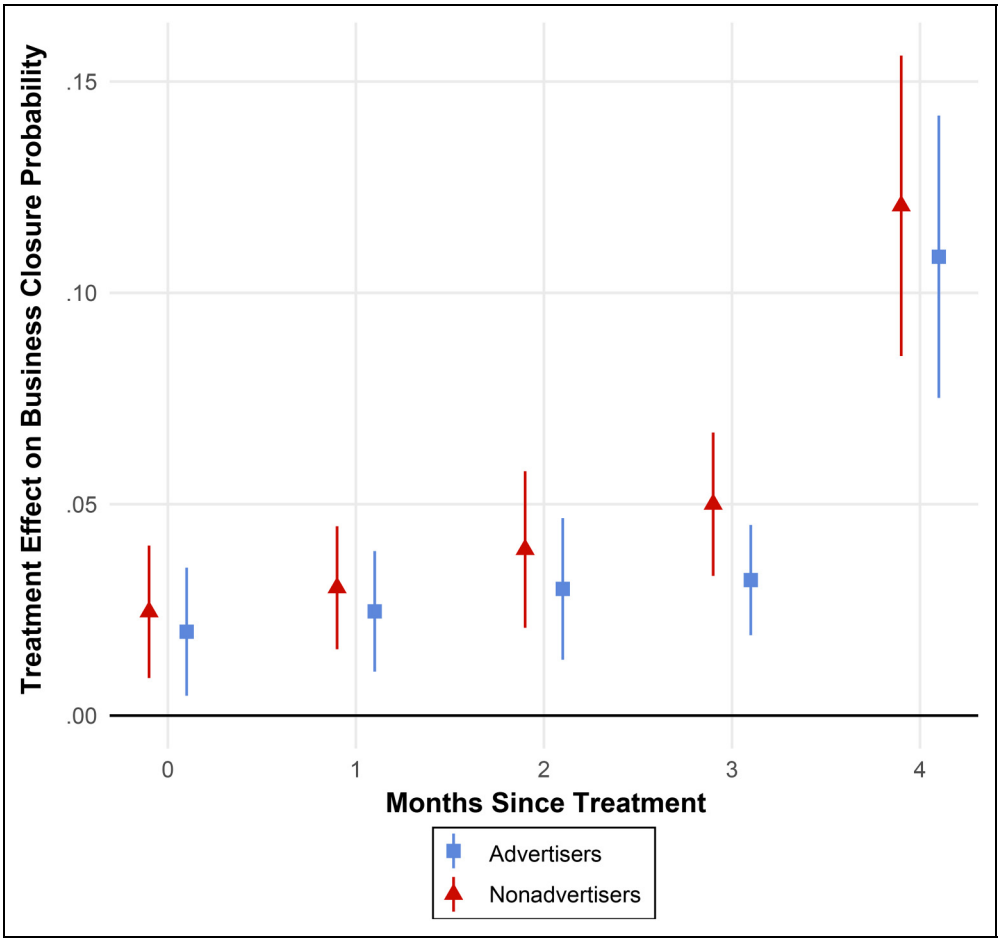


Figure 6. ATT Event-Study Estimates by Advertiser Status, Using an Alternative Stay-at-Home Order Classification (ETWFE).
Notes: A month is classified as treated if a stay-at-home order was in effect for at least two days.

average and report the ETWFE results in Table 8, Row 2. We again find that, when spending is measured using the industry average, businesses with higher ad spending continued to have lower ATT estimates than those with lower ad spending.

Overall, the results using these alternative advertising measures provide additional support for our argument that advertising helps alleviate the adverse effect of stay-at-home orders and increase business resilience.

Results by Business Size and Industry

We explore the heterogeneous effect of stay-at-home orders and advertising's moderating effect in this section. Figure 9 displays the ATT estimates broken down by business size (indicated by the number of employees). We find that advertising appears particularly effective for small businesses compared with large businesses. Note that these results are directional as the estimates are imprecise.

We also break down the ATT estimates by industry sectors, as shown in Figure 10. We again see variations in both the ATT estimates and advertising's moderating effects across industries. For example, retail and construction industries appear to have benefited more from Facebook advertising, while advertising barely made any difference in the transportation industry and even had a negative effect on the agriculture industry. Nevertheless, directionally, Facebook advertisers in all industries (except for agriculture, which accounts for around 2% of the data) have lower ATT estimates compared with their nonadvertiser counterparts.

In summary, our analyses show that, while stay-at-home orders increased business closure overall, the adverse effects were less severe for social media advertisers than for nonadvertisers. The results are robust across different model specifications and measures of advertising activities. We observed similar trends across businesses of different sizes and industries.

Conclusion

In this study, we set out to examine whether governments' stay-at-home orders affected SMB operation during the COVID-19 pandemic and the potential moderating role of social media advertising, focusing on Facebook data. Analyzing data from more than 79,000 businesses across 49 countries, we show that government lockdowns led to an

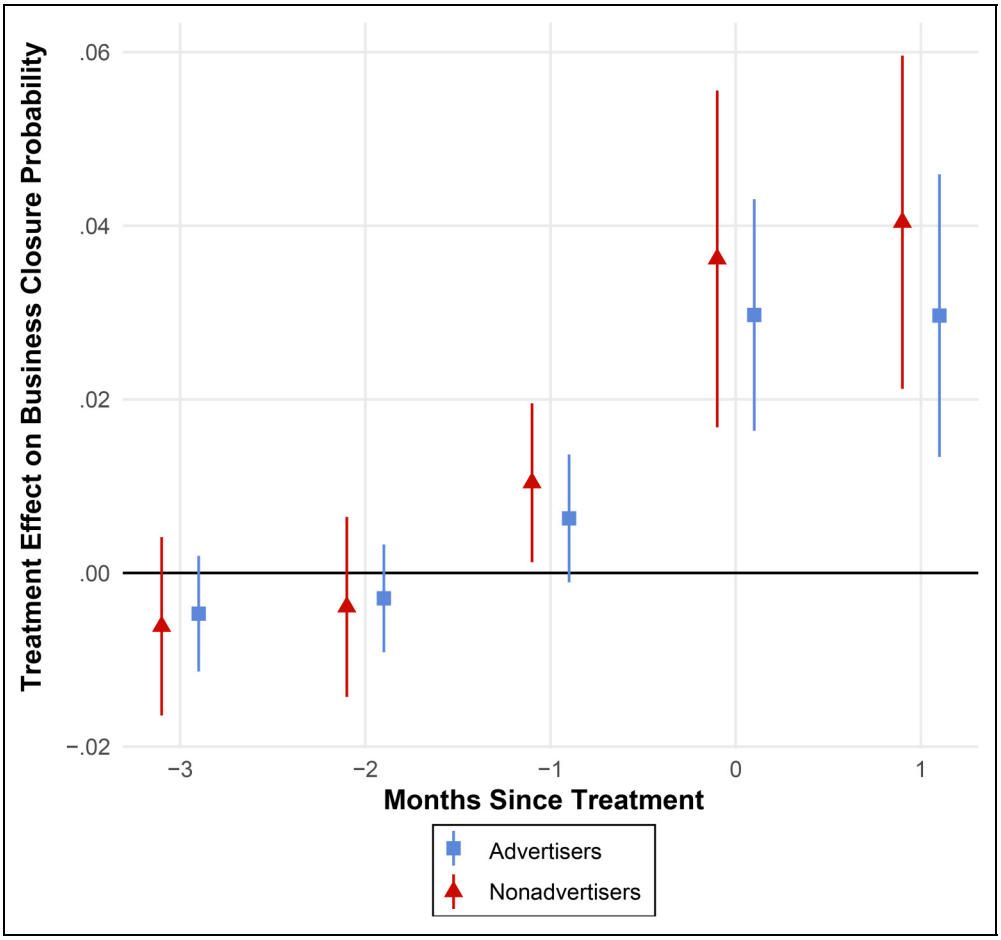


Figure 7. ATT Event-Study Estimates by Advertiser Status (Callaway and Sant'Anna DID Estimator).
Notes: Estimates are derived from the Callaway and Sant'Anna DID estimator and aggregated using balance_e = 1. A significant result one month prior to treatment suggests the presence of an anticipation effect.

Table 6. Moderating Effect of Advertising on ATT (Callaway and Sant'Anna DID Estimator).

	Overall	Advertiser	Nonadvertiser
ATT(−3)	−.005 [−.010, .000][a]	−.005 [−.011, .002]	−.006 [−.016, .004]
ATT(−2)	−.003 [−.008, .002]	−.003 [−.009, .003]	−.004 [−.014, .006]
ATT(−1)	.008 [.002, .013][a]	.006 [−.001, .014]	.010 [.001, .020][a]
ATT(0)	.032 [.022, .042][a]	.030 [.016, .043][a]	.036 [.017, .056][a]
ATT(1)	.033 [.022, .044][a]	.030 [.013, .046][a]	.040 [.021, .060][a]

[a]The confidence band does not cover 0.
Notes: The table shows Callaway and Sant'Anna DID estimates of ATT, broken down by advertiser status, controlling for business size and social media popularity (number of Facebook fans). The range in brackets represents the 95% simultaneous confidence band.

increase in business closure probability, but Facebook advertising helped attenuate this negative effect, with substantial variation in the results across business sizes and industries.

We believe our exploration contributes to the literature in several ways. First, we analyze an original and unique dataset and uncover evidence that lockdowns indeed intensified business closures *globally* and across industries (Alexander and Karger 2023; Gourinchas et al. 2025). Second, we extend the literature by investigating whether and how advertising was a useful tool for SMBs during the pandemic. Prior research has shown that advertising in general helps U.S. businesses survive during times of economic turbulence (Angulo-Ruiz et al. 2022; Jindal and McAlister 2015), but the impact of online advertising in particular on business survival is not well understood in a global context. Our findings indicate that online advertising played an important role in alleviating some of the adversity businesses faced during the pandemic. These directional findings

Table 7. Moderating Effect of Advertising on ATT (Number of Ads).

	Alternative Ad Measure by Number of Ads	
	Bottom 50%	**Top 50%**
ATT (0)	.026 (.009)**	.021 (.009)*
ATT (1)	.032 (.008)***	.025 (.008)***
ATT (2)	.042 (.010)***	.029 (.009)***
ATT (3)	.062 (.010)***	.022 (.006)***
ATT (4)	.122 (.018)***	.107 (.016)***

*$p < .05$.
**$p < .01$.
***$p < .001$.
Notes: The table shows ETWFE (Wooldridge 2021) estimates of ATT by the number of active ads, controlling for country GDP per capita, business size and social media popularity (number of Facebook fans). Top 50% advertisers are defined as those with an above-median number of active ads. Standard errors are clustered at the country level.

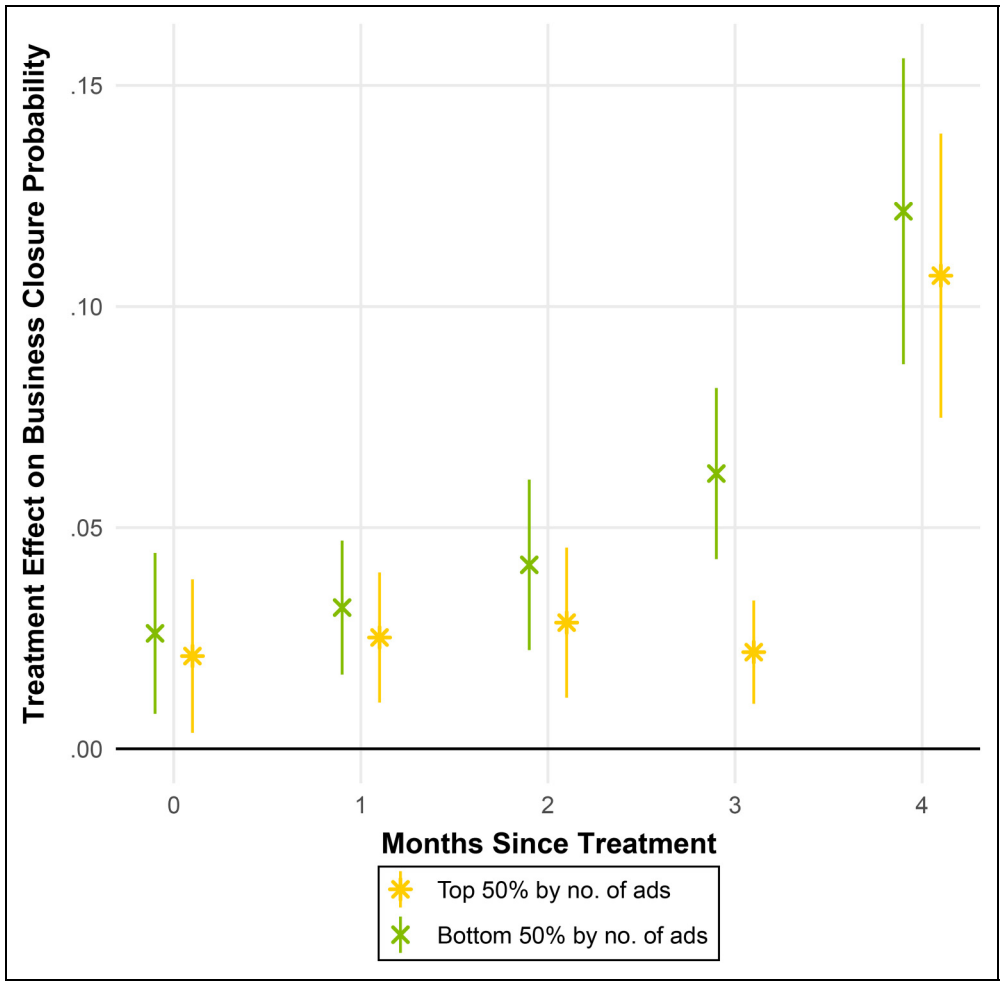


Figure 8. ATT Event-Study Estimates by Number of Active Ads (ETWFE).
Notes: Top 50% advertisers are defined as those with an above-median number of active ads.

are all the more important as reports suggest that firms have historically tended to cut back rather than increase marketing investments in times of severe economic turmoil (Özturan, Özsomer, and Pieters 2014; Rollins, Nickell, and Ennis 2014). Third, our findings are consistent with the view that advertising improves business performance through a lagged effect. Our study suggests that, directionally, advertising technology may serve as "insurance" against sudden

Table 8. ATT Estimates by Ad Spending Quintiles.

	Quintiles				
	1	2	3	4	5
1	.043 (.009)***	.031 (.009)***	.025 (.008)**	.022 (.007)**	.020 (.006)**
2	.033 (.012)**	.020 (.010)*	.025 (.011)*	.019 (.012)	.018 (.012)

*$p < .05$.
**$p < .01$.
***$p < .001$.
Notes: The table shows ETWFE (Wooldridge 2021) estimates of ATT by quintiles. Quintile 1 contains businesses with the lowest ad spending; Quintile 5 contains businesses with the highest ad spending. In Row 1, quintile categorization uses the ad spending amount only among advertisers. In Row 2, quintile categorization is applied to ad spending measured using the average spending of competitors in the same industry and country (excluding own expenditures). Treatment is the implementation of a stay-at-home order. Outcome is the treatment effect on business closure probability. Results show that businesses with the highest ad spending were affected less by the stay-at-home order, controlling for country GDP per capita, business size, and social media popularity (number of Facebook fans).

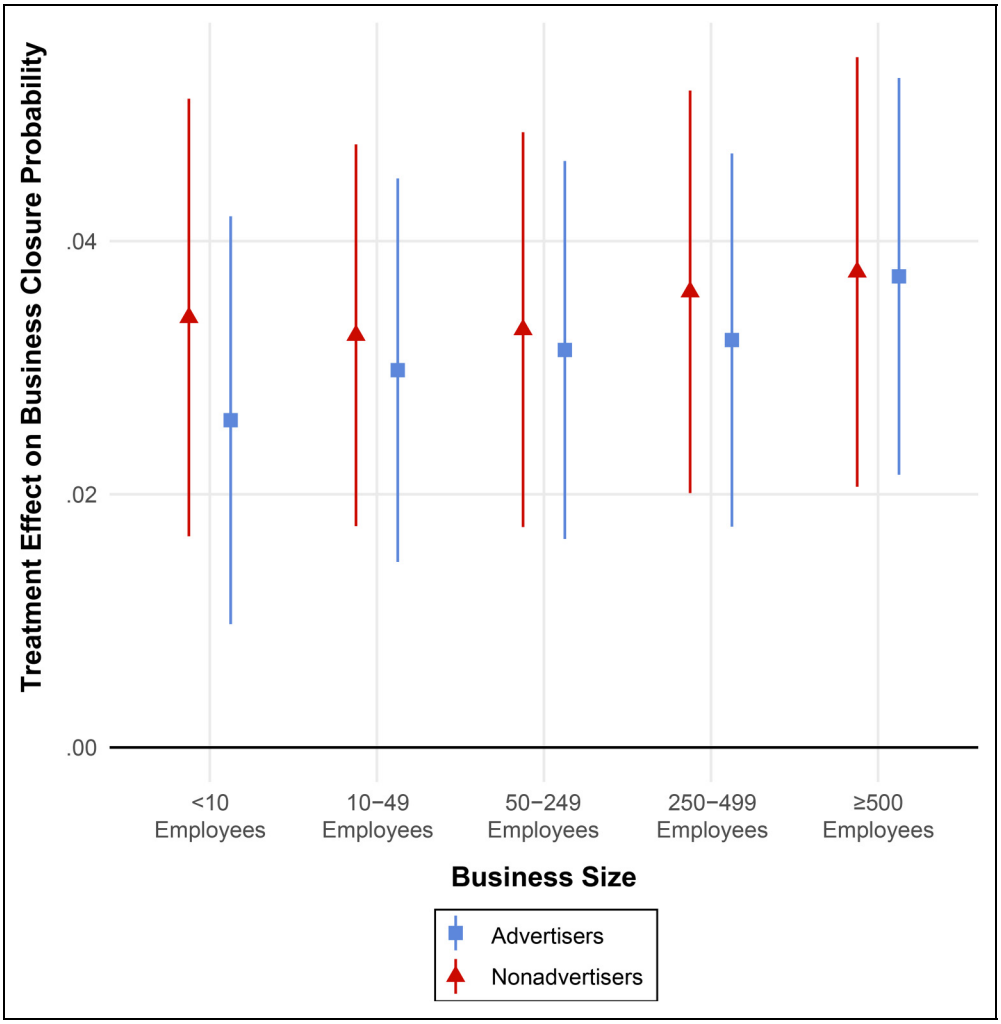


Figure 9. ATT Estimates by Business Size and Advertiser Status (ETWFE).
Notes: Size is defined by the number of employees.

and extreme demand shocks (Lewis and Reiley 2014; Manchanda et al. 2006). The findings further highlight that marketing technology can play an important role in building SMB resilience in light of severe demand shocks (Chandy et al. 2021).

Our findings provide important implications for stakeholders. For businesses, maintaining or even increasing social media advertising during times of crisis can be a counterintuitive but potentially beneficial strategy, as it helps build market-based intangible assets such as brand recognition and customer

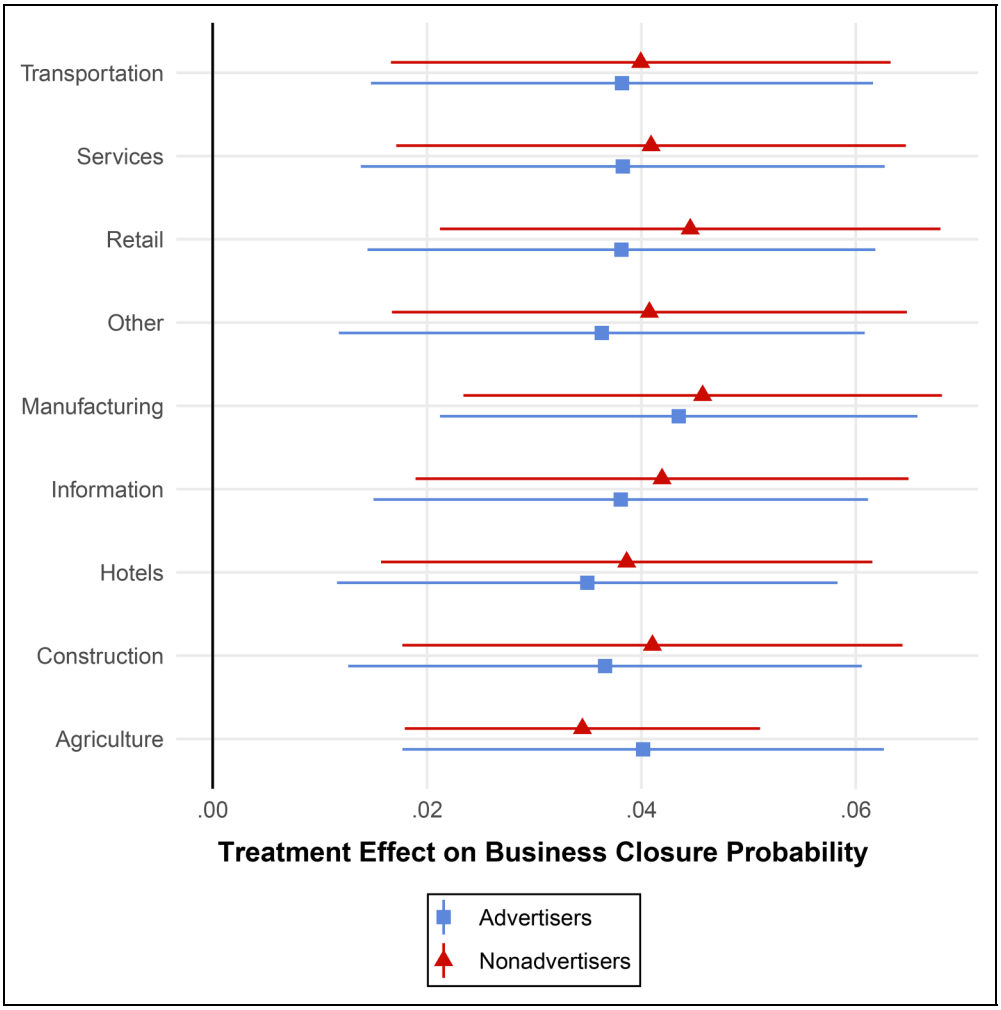


Figure 10. ATT Estimates by Industry and Advertiser Status (ETWFE).

engagement. A potential implication for policy makers is to consider subsidizing SMBs' social media advertising campaigns, especially during economic downturns. Financial institutions in turn could consider providing credit lines or loans to assist SMBs in sustaining their social media advertising campaigns during financial hardships.

Our study has several limitations, mostly originating from the correlational nature of our findings. First, in addition to stay-at-home orders, governments around the world also implemented different economic and health policies during the pandemic, which we did not take into account in the current research. Future research should examine the effect of different policies on business operations. Second, we used Facebook advertising as the sole source of online advertising data. While most SMBs are likely to rely on prominent digital advertising platforms such as Facebook as their main advertising channels, we cannot rule out the possibility that there might have been other unobserved advertising activities. Third, we did not observe and thus could not control for key business characteristics such as financial health and age. Furthermore, businesses' advertising decisions are inherently endogenous and depend on their own performance as well as that of their competitors, leading our analyses to be correlational in nature. We encourage future research to carefully address the endogeneity concern.

Nevertheless, we believe our research raises important questions that have potential to enrich the scientific discourse (Golder et al. 2023). The first question is whether our findings apply to other modes of online advertising. The predictions provided by our analysis are not limited to the use of Facebook advertising, and we expect that different modes of online advertising, such as banner ads and search advertising, might also have supported the performance and survival of businesses during the pandemic. The second question is whether online advertising technology was especially relevant during the pandemic in light of the increase in internet usage and e-commerce expansion. More fully understanding how different modes of advertising interacted with the increased internet usage during the pandemic is important. While our study focuses on Facebook advertising, gaining insights into SMBs' overall advertising spending and allocation across different media and channels would shed further light on the role of online

advertising in enhancing business resilience—including potential interaction effects between different media. The third question is whether specific targeting goals and/or strategies in social media advertising could further alleviate the negative effects of stay-at-home orders. As social media platforms have the ability to tailor the ad display to suitable users, it is possible that a carefully designed targeting strategy could amplify the moderating effect, which future research can investigate. Finally, as the odds of the next pandemic occurring are likely to increase (*Bloomberg News* 2023), it is imperative that we deepen our understanding of how social media advertising can most effectively enhance small business resilience.

Editor

Beth Fossen

Declaration of Conflicting Interests

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